Lesson 05: Research Questions and Hypotheses

English Stage 4 (Psychology) · Unit 1: Psychology as a Science

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 2.4 Generating Good Research Questions; 2.5 Developing a Hypothesis.

In this lesson

You will learn how to turn a research idea into an empirically testable question, how to judge whether a question is interesting and feasible, and how theories lead to hypotheses. On the language side, you will practice writing hypotheses with *if … then … should* and turning them into questions. You will then write the research question and hypothesis of your proposal.

Learning goals

After this lesson you should be able to:

- Generate empirically testable questions about a variable or a relationship between variables.
- Evaluate a research question for interestingness and feasibility.
- Distinguish a theory from a hypothesis and describe the hypothetico-deductive method.
- Write a hypothesis that is testable, falsifiable, logical and positive.

Key language

Term or phrase	Meaning	Example
variable	something that varies across people or situations (more in Lesson 06)	talkativeness; family size
empirically testable research question	a question expressed in terms of a single variable or a relationship between variables	Is there a relationship between family size and talkativeness?
discussion	the final major section of a research article: a summary, implications, limitations and suggestions for future research	Its suggestions can become your research question.
interestingness	a property of research questions based on whether the answer is in doubt, fills a gap in the research literature and has important practical implications	Note taking by hand: interesting for education policy
feasibility	the extent to which a research question can be answered with available resources	time, money, equipment, skills, participants
theory	a coherent explanation or interpretation of one or more phenomena	Zajonc's drive theory

Term or phrase	Meaning	Example
hypothesis	a prediction about a new phenomenon that would be observed if a particular theory were true	Cockroaches should run faster when others are present.
hypothetico-deductive method	the general way researchers use theories to generate new research and, in the process, test and revise the theories	theory, hypothesis, study, revised theory
deductive reasoning	reasoning from a broad, general theory to a more specific hypothesis	from drive theory to a prediction about cockroaches
inductive reasoning	reasoning from specific observations or findings to a more general hypothesis	used when no theory addresses the question
falsifiable	there is an observation that would count as evidence against the claim	Cockroaches running the easy runway more slowly with an audience would count against the hypothesis.
to the extent that	in proportion to how much	"a research question is interesting to the extent that its answer is in doubt"
it is generally good practice to	it is usually the right thing to do	"... to use methods that have already been used successfully"

Rules and explanations

The ideas

From idea to empirically testable question (2.4)

- Read the **discussion** section of a recent article: researchers often suggest specific questions for future research, already judged interesting by experts.
- Or generate your own. First treat the behavior as a variable and ask how frequent or intense it is (How many words do people speak per day?).
- If that has been answered, ask about a relationship: What are possible **causes** and **effects** of the behavior? What **types of people** and what **situations** might show more or less of it? Each answer is a second variable: Is there a relationship between family size and talkativeness? Are people more talkative in same-sex groups?
- If the relationship has also been studied, do not give up (its publication shows the scientific community is interested): refine it. Are there other ways to define and measure the variables (the number of *different people* spoken to each day)? Types of people or situations where the relationship is stronger or weaker (elderly people, other cultures)?

Evaluating questions: interestingness (to the scientific community, not just to you)

Factor	Test	Example
The answer is in doubt	Can you think of reasons to expect at least two different answers?	Talkativeness: the stereotype suggests yes, similar verbal abilities suggest no. A punch in the jaw hurts: only one possible answer.
It fills a gap	It is not yet answered, and it is a natural question for people who know the literature.	Note taking by hand, given research on shallow processing
It has practical implications	The answer matters for real decisions.	classroom technology policies; laws on cell phones and driving

Feasibility depends on time, money, equipment and materials, technical knowledge and skill, and access to participants. Research does not have to be complicated to be important: a convenience sample of students and a paper-and-pencil task can be enough. Use methods that others have used successfully: they are "tried and true", and your results can be compared with theirs.

Theories and hypotheses (2.5)

- A **theory** goes beyond the phenomena it explains by adding things not directly observed. Zajonc's **drive theory**: being watched by others creates physiological arousal, which increases the dominant (most likely) response. So an audience improves performance on well-practiced tasks (social facilitation) and worsens it on unpracticed tasks (social inhibition).
- In everyday speech, a "theory" may be a wild guess; in science, a theory can be extensively tested and accepted, like evolution by natural selection or the germ theory of disease.
- A **hypothesis** is a specific prediction, often but not always derived from a theory. Theory and hypothesis have an *if-then* relationship: "*If* drive theory is correct, *then* cockroaches should run through a straight runway faster, and a branching runway more slowly, when other cockroaches are present."

Deriving hypotheses

1. Ask whether a theory implies an answer to your question. If habituation theory is correct (expressive writing helps because people get used to negative thoughts and feelings), writing about *positive* experiences should not improve health.
2. Focus on a part of the theory not yet observed: people should show fewer signs of distress with each writing session.
3. Best of all, test **competing theories**. Schwarz and colleagues (1991) asked people to recall 6 (easy) or 12 (difficult) examples of their own assertiveness. The number-of-examples theory predicts that the 12-example group will judge itself more assertive; the ease-of-retrieval theory predicts the 6-example group. Result: the 6-example group did, supporting ease of retrieval.

The hypothetico-deductive method. Researchers construct or choose a theory, derive a hypothesis, test it in a study, and reevaluate the theory, revising it if necessary; then the cycle repeats. Zajonc built drive theory from contradictory findings and tested it with cockroaches (Zajonc, Heingartner, & Herman, 1969): with an "audience" of other cockroaches, they ran a straight runway (easy) faster but a cross-shaped maze (difficult) more slowly, as predicted.

A good hypothesis is:

- **testable and falsifiable**: evidence could show it is false (Popper);
- **logical**: based on theory or observations, by deductive or inductive reasoning, not a random guess;
- **positive**: it states that a relationship or effect exists, not that it does not (the reason lies in statistics, Unit 4).

Language focus

1. Writing hypotheses. Hypotheses use *should* (= is expected to) and comparisons:

- *If* + theory + *is correct, then* + prediction with *should*: "If drive theory is correct, then cockroaches **should** run through a straight runway **faster** ... when other cockroaches are present."
- Group predictions: "participants who recalled six examples **should judge** themselves as more assertive"
- A hypothesis can be rephrased as a question: "**Do** cockroaches run through a straight runway faster when other cockroaches are present?"

2. Evaluating with degree and possibility.

- "If you can think of reasons to expect at least two different answers, then the question **might be** interesting. If you can think of reasons to expect only one answer, then it **probably is not**."
- "a research question is interesting **to the extent that** its answer is in doubt"

> **Note for Arabic speakers** - Learn the irregular plurals: hypothesis, hypotheses; phenomenon, phenomena; criterion, criteria ("two criteria for evaluating research questions"); analysis, analyses. - In a hypothesis, *should* means "is expected to", not "must": "Students who nap should recall more."

Practice

1. Reading questions

1. Why can the discussion section of a recent article be a good source of research questions? (2.4)
2. Which four general questions turn a behavior into a question about a relationship? Give the book's talkativeness example.
3. If your question has already been studied, should you give up? Explain with the book's reasons.
4. Why is "Do people feel pain when you punch them in the jaw?" not interesting, while "Are women more talkative than men?" is?
5. In Schwarz et al.'s study, why did the result count as convincing evidence for one theory over the other?
6. Describe Zajonc's cockroach study: the tasks, the conditions and the results.
7. Name the three characteristics of a good hypothesis.

2. Book exercises. Sections 2.4 and 2.5 have no exercises of their own; Chapter 2's exercises are in Section 2.9 (Lesson 06).

3. Language exercise

A. Rewrite as a question: "People should show fewer signs of emotional distress with each new writing session."

B. Rewrite as an if-then hypothesis with *should*: "Is there a relationship between family size and talkativeness?" (Give a reason in the *if*-clause.)

C. The book says hypotheses should be positive. Rewrite each one positively.

1. There is no difference in the number of words that women and men speak per day.
2. If habituation theory is correct, expressive writing about positive experiences should not improve health.

Speak with your tutor

- "My research question is: [question]. Ask me the four general questions (causes, effects, types of people, situations) to generate alternatives."
- "Challenge the interestingness of my question: give me reasons why the answer might not be in doubt. I will defend it."
- "PSYC 110 Lecture 10 (Evolution and Rationality) describes the availability bias: we judge how frequent something is by how easily examples come to mind. The book's reference to Schwarz et al. is subtitled 'Another look at the availability heuristic'. Ask me how the two ideas are related."
- "Act as a reviewer. Tell me whether my hypothesis is testable, falsifiable, logical and positive."

Writing task

Research proposal, part 2: research question and hypothesis. Write 250 to 350 words. Start from your Lesson 04 topic.

Include:

1. **Your research question**, expressed in terms of variables, and how you reached it from your topic (which general question you used, how you refined it).
2. **An evaluation** of its interestingness and feasibility (you are planning a study, not running it).
3. **The theory or observation** behind your hypothesis, and whether you reasoned deductively or inductively.
4. **Your hypothesis**, in if-then form and as a plain prediction.

Example for the default topic (sleep and memory in university students). The Stage 3 question "How does the amount of sleep affect memory and learning in university students?" becomes the empirically testable question "Do students who sleep after learning remember more than students who stay awake for the same length of time?" Theory: the idea, mentioned in PSYC 110 Lecture 17, that memories are consolidated during sleep (check it in the literature from Lesson 04). Hypothesis: "If sleep helps to consolidate new memories, then students who take a nap after learning a list of word pairs should recall more pairs than students who stay awake for the same time." A nap (for example, a nap

opportunity of up to 90 minutes, with the other group resting quietly awake for the same time) is more feasible, and kinder to participants, than a night without sleep.

If you work on your own topic, follow the same four steps.

Self-editing checklist:

- Is my question about a variable or a relationship between variables?
- Can I give reasons to expect at least two different answers?
- Did I consider every feasibility factor?
- Did I name the theory or observations behind my hypothesis?
- Can I describe a result that would show my hypothesis is false?
- Is my hypothesis positive (it predicts an effect)?
- Did I use *if … then … should* and plurals such as *hypotheses* and *criteria* correctly?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- PSYC 110 Lecture 17 (sleep and dreams in Chapters 6 and 7, from 38:09): https://oyc.yale.edu/psychology/psyc-110/lecture-17
- PSYC 110 Lecture 10 (availability bias and confirmation bias in Chapter 5, from 39:48): https://oyc.yale.edu/psychology/psyc-110/lecture-10

Answer key

1. Reading questions

1. Researchers there suggest directions for future research, often as specific questions, and these questions have probably already been judged interesting and important by experienced researchers (2.4, first paragraph).
2. What are possible causes of the behavior? What are possible effects? What types of people might show more or less of it? What types of situations might elicit more or less of it? Talkativeness: family size as a possible cause; more talk in same-sex groups than in mixed-sex groups (2.4).
3. No. Its publication shows that the scientific community is interested, and the question can almost certainly be refined: other ways to define and measure the variables, and types of people or situations in which the relationship might be stronger or weaker (2.4).
4. There is no reason to expect any answer but "yes", so the answer is not in doubt. For talkativeness there are reasons to expect both answers: the stereotype suggests yes, but women's and men's verbal abilities are fairly similar (2.4, "Interestingness").
5. The two theories made opposite predictions, so only one could be confirmed. Participants who recalled fewer (six) examples judged themselves more assertive, which supports the ease-of-retrieval theory (2.5).
6. Cockroaches escaped from light into a dark chamber, either down a straight runway (easy) or through a cross-shaped maze (difficult), alone or with other cockroaches watching from clear "audience boxes". With an audience, they reached the goal faster in the runway and more slowly in the maze, confirming the hypothesis and supporting drive theory (2.5, "Theory Testing").
7. Testable and falsifiable; logical; positive (2.5, last paragraph).

2. Book exercises. See Lesson 06.

3. Language exercise

A. Do people show fewer signs of emotional distress with each new writing session?

B. (example answer) "If people become more talkative through daily practice in conversation, then people from larger families should be more talkative than people from smaller families."

C. (example answers) 1. Women speak more words per day than men. (A positive hypothesis can still be wrong: Mehl's data did not support this one.) 2. If habituation theory is correct, expressive writing about traumatic experiences should improve health more than expressive writing about positive experiences.

My notes